

RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

5. Microwave Generators

5 hrs · asked in **23 of 24 papers** · 5 syllabus sub-topics · **200 marks, 9.7% of the archive** → two tubes (klystron, magnetron) answer most of it; one idea (bunching) is the 2-mark rider on almost every question.

READ THIS FIRST

Chapter 5 is descriptive, with no numerical PYQ in 24 papers, and the questions come in a fixed shape: **a 2-mark definition, then a 6 to 8 mark “construction and working”**.

1. **The 2-mark riders** (bunching, density modulation, transit time, cross-field effect, slow-wave structure) → each has a boxed two-line answer below. Bunching alone is asked in 9 papers.
2. **Klystrons** (two-cavity, multicavity, reflex; 9 papers) → §5.3 and §5.4. One figure (the Applegate diagram) explains all three.
3. **Magnetron** (9 papers) → §5.5. The “45° / 60° / 90° phase shift between adjacent cavities” versions are the same answer plus one formula, $\phi = 2\pi n/N$.

Two things worth knowing before the paper starts:

- Every tube here does the same three things: **velocity-modulate** a beam, let it **bunch**, and **decelerate the bunches** in a field that takes their energy. Say this in the first line of any tube answer and the structure of the answer follows.
- Linear-beam (O-type) tubes: klystron, reflex klystron, TWT, BWO. Crossed-field (M-type) tubes: magnetron, crossed-field amplifier. An examiner who writes “field effect” or “cross-field” wants the M-type family.

5.1 Conventional Tubes at Microwaves and the Transit-Time Effect

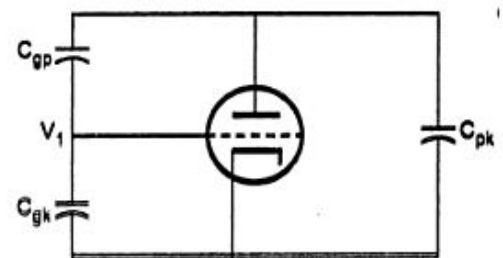
How is the output of conventional tubes reduced at microwaves? What is transit-time effect? **2 2+6**

2+8

HOT 3/24 70 Ma, 77 Ch, 79 Bh · always the 2-mark opener to a tube question

Why triodes and pentodes fail above about 1 GHz

- **Inter-electrode capacitance** C_{gp}, C_{gk}, C_{pk} :
 - Reactance $1/\omega C$ falls with frequency → shorts the input and output → gain falls.
 - Cure (spread electrodes, smaller area) raises transit time: a **compromise**.
- **Lead inductance**: ωL grows → degenerative feedback; cure: short leads (disk-seal tubes).
- **Transit-time effect** (below).
- **Cathode emission**: more current needs a bigger cathode, which adds capacitance; filament voltage and temperature are limited.
- **Power losses**: skin-effect resistance grows as $\sqrt{f}$; dielectric loss in the glass envelope; radiation from leads.



Inter-electrode capacitances of a triode. Karkee, Ch-5 deck, slide 4.

Transit-time effect: the 2-mark answer 2

- **Transit time** = time an electron takes from cathode to anode. **Transit-time effect**: when this becomes a significant fraction of one RF cycle, the grid voltage changes while the electron is in flight → plate current lags the grid voltage → efficiency and gain collapse.
- Numbers: $\tau = 1 \text{ ns}$ is 0.001 cycle at 1 MHz but a **whole cycle** at 1 GHz. Above about 0.1 cycle the loss is serious.
- Estimate: 100 V gives $v = \sqrt{2eV/m} = 5.93 \times 10^6 \text{ m/s}$; a 1 mm gap then takes $3.4 \times 10^{-10} \text{ s}$, i.e. 0.34 cycle at 1 GHz.
- Cure in a conventional tube: closer electrodes and higher voltage (fights the capacitance). **Cure in a microwave tube: exploit it.** Klystrons, TWTs and magnetrons turn transit time into velocity modulation and bunching.

verified/added **The four effects solid-state generators exploit**

Effect	What happens	Device
Transit time	Carrier drift across a region delays current behind voltage; a delay near half a cycle gives negative resistance	IMPATT, BARITT
Tunnelling	Carriers pass through a thin potential barrier they could not climb classically; current falls as bias rises past the peak	Tunnel diode
Avalanche	Strong field accelerates carriers until impact ionization knocks electrons out of covalent bonds, a chain reaction	IMPATT (with transit time)
Transferred electron (Gunn)	Above a threshold field (about 3.2 kV/cm in GaAs) electrons move to a low-mobility upper valley: negative differential mobility, no p-n junction	Gunn diode, §4.11

5.2 Velocity Modulation, Density Modulation and Bunching**What is bunching effect (density modulation)? Explain velocity modulation and bunching** 2 5 10

TOP 9/24 70 Bh, 71 Bh, 72 Ash, 72 Ma, 73 Ma, 75 Bh, 80 Ba, 80 Bh, 81 Bh · the most repeated ask in the chapter

2-mark answer. Velocity modulation: an RF voltage across a gap speeds up electrons crossing in one half-cycle and slows those crossing in the other. **Bunching (= density modulation)**: in the field-free drift space beyond, fast electrons catch up with slow ones sent earlier, so the uniform beam gathers into **bunches** once per RF cycle. The beam current is now modulated at the RF frequency and can drive a second cavity.

The equations (for the 5 and 10 mark versions)

- Beam accelerated by V_0 : $v_0 = \sqrt{2eV_0/m} = 0.593 \times 10^6 \sqrt{V_0} \text{ m/s}$.
- Buncher gap voltage $V_1 \sin \omega t$. Leaving at t_1 :

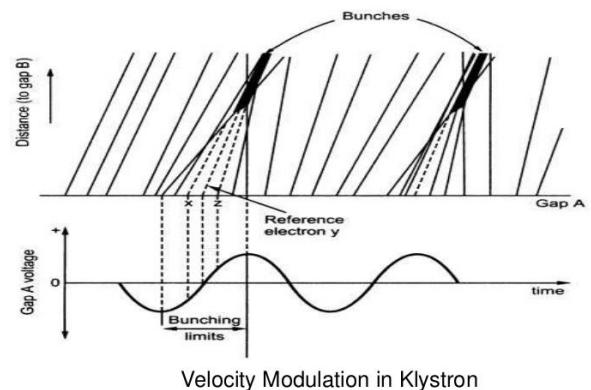
$$v(t_1) = v_0 \left[1 + \frac{\beta_i V_1}{2V_0} \sin \omega t_1 \right]$$

$\beta_i = \frac{\sin(\theta_g/2)}{\theta_g/2}$, the **beam coupling coefficient**; $\theta_g = \omega d/v_0$ the gap transit angle.

- Three reference electrons: #2 crosses at zero voltage (unchanged), #3 on the rising half (accelerated), #1 before it (decelerated). #1 and #3 converge on #2.
- Arrival at the catcher, drift length L : $t_2 = t_1 + L/v(t_1)$.
- **Bunching parameter**

$$X = \frac{\beta_i V_1}{2V_0} \theta_0, \quad \theta_0 = \frac{\omega L}{v_0}$$

- Catcher current: $i_2 = I_0 + \sum_n 2I_0 J_n(nX) \cos n\omega(t_2 - \tau_0)$.
- Fundamental $2I_0 J_1(X)$ peaks at **$X = 1.841$** , where it is $1.164 I_0$ → optimum drift length.

Two cavity klystron

Velocity Modulation in Klystron

Applegate diagram: each line is one electron's distance vs time; steep lines are fast. Lines converge (a bunch) around the electron that crossed at zero gap voltage. Karkee deck, PDF p22.

- **Checked by simulation in tubes.py**: electrons launched with this velocity law and drifted ballistically give a fundamental of exactly $2J_1(X)$.
- Too short a drift ($X < 1.84$): weak bunches. Too long: electrons overtake and the bunch spreads again (**debunching**).

5.3 Two-Cavity and Multicavity Klystrons

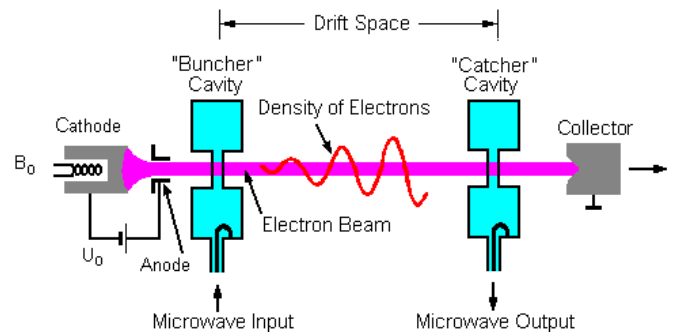
Describe the construction and operation of a two-cavity / multicavity klystron amplifier or oscillator

4 5 2+7 2+8

TOP 7/24 69 Bh, 70 Ma, 71 Bh, 72 Ash, 78 Ch, 80 Ch, 82 Ba · 70 Ma two-cavity; the rest mostly multicavity

Construction (two-cavity amplifier)

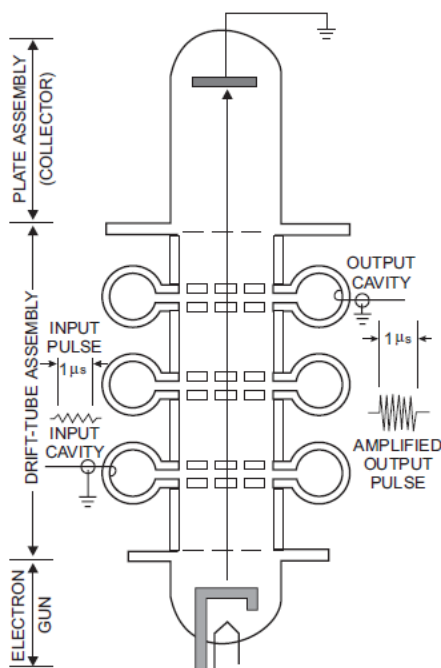
- **Electron gun:** heated cathode, focusing electrode, anode at $+V_0$ forms a narrow beam; an axial magnetic field keeps it focused.
- **Buncher cavity (input):** re-entrant cavity with grids across the beam; RF in via coaxial loop.
- **Drift space:** field-free tube of length L .
- **Catcher cavity (output):** identical cavity, RF out via loop.
- **Collector:** absorbs the spent beam.
- Cavities and collector at the same dc potential, so the beam drifts at constant mean speed.



Two-cavity klystron. Karkee deck, slide 14.

Operation

1. Weak RF in the buncher sets up $V_1 \sin \omega t$ across its gap.
 2. **Velocity modulation:** electrons crossing are accelerated or decelerated (§5.2).
 3. **Bunching** in the drift space: velocity modulation becomes density modulation.
 4. Bunches reach the catcher once per cycle and **induce** RF current in it. Its field builds up to **oppose** the bunches: they are decelerated and give up kinetic energy, while the thin stream between bunches is accelerated and takes back only a little.
 5. Net energy flows from beam to RF $\rightarrow$ amplified output. The beam leaves slower and is collected.
- **Correction to the source** (PDF p23): it says "all the kinetic energy is converted". Only part is; the beam still reaches the collector with most of its speed. Theoretical maximum efficiency is $\frac{1}{2} \times 1.164 = 58\%$ (at $X = 1.841$, catcher voltage equal to V_0); practical about 40%. verified/added
 - Gain of two cavities is low (about 10 dB) $\rightarrow$ multicavity.



Multicavity klystron. Karkee deck, slide 19.

Multicavity klystron (amplifier)

- One or more **intermediate cavities** between buncher and catcher, not externally driven.
- Weak bunches from cavity 1 excite cavity 2 $\rightarrow$ its larger gap voltage re-modulates the beam $\rightarrow$ tighter bunches $\rightarrow$ and so on.
- Each stage adds gain: **30 to 70 dB** overall.
- **Stagger tuning** the cavities slightly apart widens the bandwidth.
- Higher efficiency and power: up to MW pulsed, kW CW.
- Uses: UHF TV and troposcatter transmitters, satellite ground stations, radar transmitters, linear accelerators.

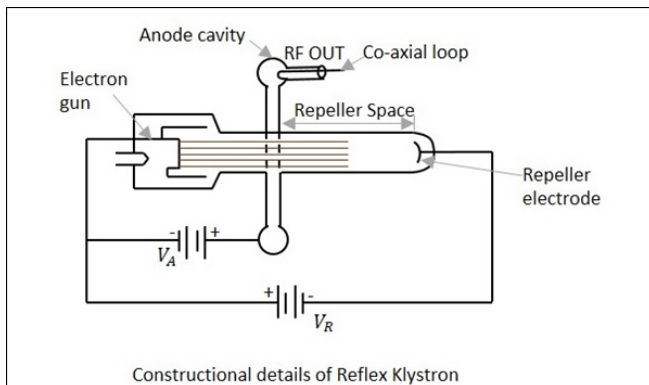
Klystron oscillator

- Feed part of the catcher output **back** to the buncher with the right phase (feedback loop, slide 18) $\rightarrow$ self-sustained oscillation. Multicavity oscillator: the same loop around a multicavity tube (72 Ash).
- Tuned by the cavities; for a single-cavity, voltage-tuned oscillator use the reflex klystron (§5.4).

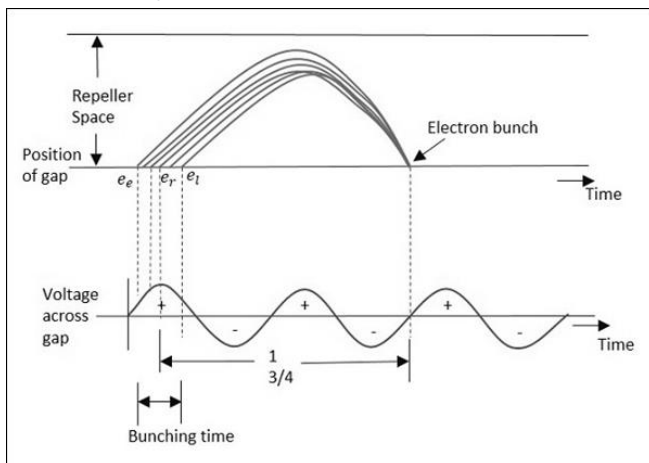
5.4 Reflex Klystron

Explain velocity modulation and bunching in a reflex klystron with neat diagrams and equations 5 10

PIN 2/24 72 Ma, 75 Bh · 75 Bh prints “multicavity reflex klystron”; a reflex klystron has one cavity



Karkee deck, PDF p29.



Electrons e_e , e_r , e_l return together: the bunch arrives $n + \frac{3}{4}$ cycles later. PDF p30.

- **Electronic tuning:** changing V_R changes T' and pulls the frequency a few percent; modes are the different n .
- **Efficiency:** maximum $\frac{2X'J_1(X')}{2\pi n + 3\pi/2}$ with $X' = 2.405$: 53% for $n = 0$ (not practical), **22.7% for $n = 1$** . The deck's 22.78% is this, with X' rounded to 2.408. Practical 10 to 20%.
- **Performance:** 4 to 200 GHz, 1 mW to 2.5 W. **Uses:** local oscillator in receivers, signal source for bench tests, portable links, pump for parametric amplifiers.

5.5 Cavity Magnetron and the Cross-Field Effect

Describe the construction and working of a cavity magnetron; magnetron with 45 / 60 / 90 degree phase shift between adjacent cavities 5 2+6 8 2+8 10

TOP 9/24 70 Bh, 71 Ma, 72 Ma, 74 Ma, 77 Ch, 79 Ch, 80 Ba, 81 Ba, 81 Bh · 81 Bh's "high-power oscillator using bunching and field effects" is this tube

2-mark answer: cross-field effect (72 Ma, 78 Ch, 81 Bh). An electron in **perpendicular** dc electric and magnetic fields feels $\vec{F} = -e(\vec{E} + \vec{v} \times \vec{B})$. It does not go straight to the anode: it follows a **cycloidal** path and drifts at $v = E/B$, perpendicular to both fields. Crossed-field (M-type) tubes use this drift to keep electrons in step with an RF wave for a long interaction.

Construction

- Electron gun, **one** re-entrant anode cavity at $+V_0$, and a **repeller** electrode at $-V_R$ beyond it.
- Output via a coaxial loop from the same cavity.

Operation

1. Noise starts a small RF voltage across the cavity gap.
2. Electrons crossing it are **velocity-modulated**.
3. In the retarding repeller field they stop and turn back. A **faster** electron goes deeper and takes **longer**: round trip

$$T' = \frac{2mLv}{e(V_R + V_0)}$$
4. So an early (fast) electron, the reference and a late (slow) one return **together**: the bunch forms in the **repeller space**.
5. The bunch re-crosses the gap. If the gap field then opposes it, it gives energy to the cavity and oscillation grows.

Condition for oscillation

- The bunch forms round the electron that left at zero gap voltage (going positive), and must return when the field most strongly retards it:

$$T' = \left(n + \frac{3}{4}\right) \text{ cycles}, \quad n = 0, 1, 2$$

- Power returned to the gap $\propto -J_1(X) \sin \theta_0$: positive for θ_0 in the second half-cycle, largest at $\frac{3}{4}$. `tubes.py` scans θ_0 with simulated electrons and finds the peak at 1.750 cycles.

Construction

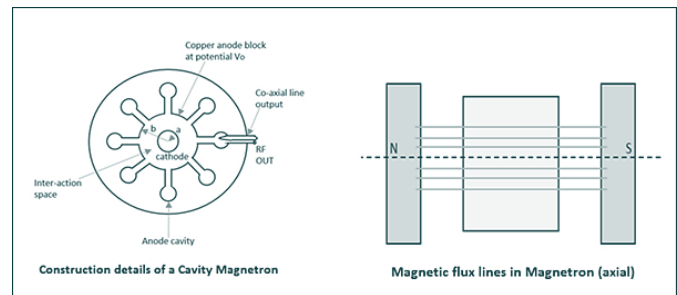
- Thick cylindrical **cathode** on the axis (radius a).
- Copper **anode block** (radius b) with N (typically 8) **resonant cavities** cut into it, opening onto the **interaction space**.
- Radial** dc field from anode voltage V_0 ; **axial** magnetic field from a permanent magnet, so $E \perp B$.
- Output: coupling loop in one cavity to a coaxial line or waveguide.
- Straps** join alternate anode segments to hold the tube in the π -mode.

Electron paths and the Hull cut-off

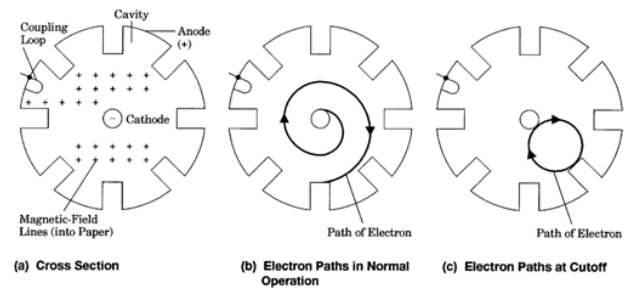
- $B = 0$: straight radial path to the anode.
- Small B : curved path, still reaches the anode.
- $B = B_c$: path just **grazes** the anode (cut-off).
- $B > B_c$: electron returns to the cathode without collecting.
- Hull cut-off field**:

$$B_c = \frac{\sqrt{8mV_0/e}}{b(1 - a^2/b^2)}$$

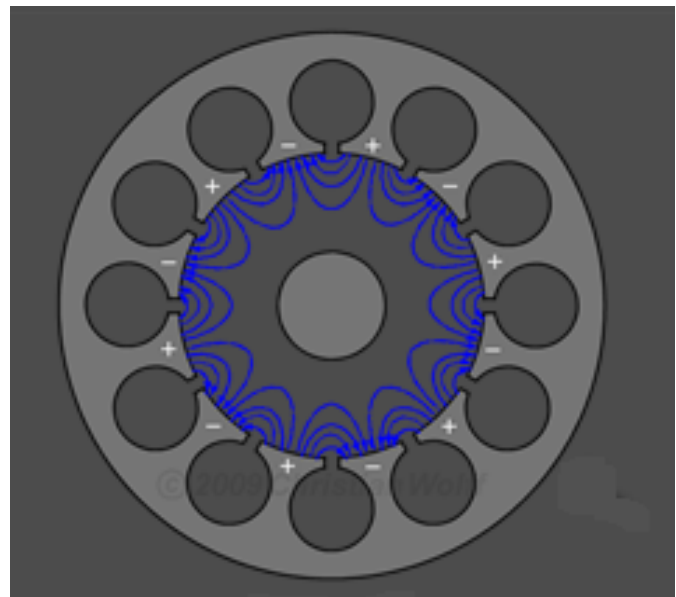
- Checked by integrating electron trajectories in `tubes.py`: for $V_0 = 10$ kV, $a = 5$ mm, $b = 10$ mm the grazing field is 0.0899 T, equal to the formula. verified/added
- The magnetron works just **above** B_c : without RF no electron reaches the anode.



Cavity magnetron cross-section and axial flux. PDF p55.



(b) normal operation, (c) at cut-off. PDF p61.



π -mode spoke pattern. Deck slide 50.

Operation (oscillator)

- Noise excites the cavities; RF fringing fields appear at the vane tips, alternating round the anode.
- Electrons circling the cathode meet these fields. Those in a **retarding** RF field slow down, feel less $v \times B$ force, and **drift outwards**, giving energy to the RF on the way.
- Those in an **accelerating** field curl back to the cathode (back-bombardment heats it, often enough to switch the heater off).
- The result is **phase focusing**: the electron cloud forms rotating **spokes**, each sitting in a retarding field. This is the magnetron's **bunching**.
- On the next half-cycle the field pattern reverses and the spokes rotate one cavity on, staying in step: sustained oscillation. Efficiency 40 to 70%.

The phase-shift versions: 45, 60, 90 degrees between adjacent cavities 6+2 8 3+7

74 Ma, 77 Ch, 80 Ba (45°); 81 Ba (60°); 72 Ma (90°)

- Around a ring of N cavities the total phase must be a whole number of turns:

$$\phi = \frac{2\pi n}{N}, \quad n = 0, \pm 1, \pm 2 \text{ up to } \pm \frac{N}{2}$$

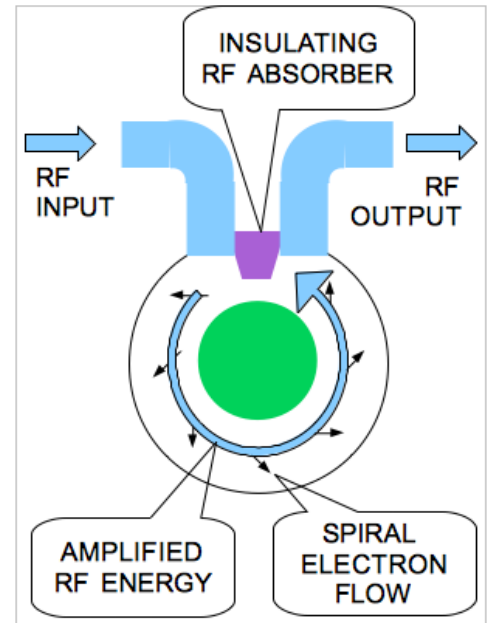
- $n = N/2$: $\phi = 180^\circ$, the **π -mode**, normal operation.
 $n = 0$: zero mode, no RF between anode and cathode, unused.
- The RF pattern repeats every $360^\circ/\phi$ cavities, so there is **one spoke** per that many cavities:
 - 45° : 8 cavities per wavelength (in an 8-cavity tube $n = 1$, one spoke).
 - 60° : 6 cavities per wavelength ($N = 6$, $n = 1$).
 - 90° : 4 cavities per wavelength ($N = 8$ gives $n = 2$, two spokes).
 - 180° : 2 cavities, $N/2$ spokes.
- **Synchronism**: the spokes rotate at the RF pattern's angular velocity ω/n ; set V_0 and B so the electron drift (E/B) matches it.
- **As a power amplifier** (80 Ba, 72 Ma): break the ring, feed RF in at one cavity and take it out several cavities later. The phase shift per cavity sets the wave's speed along the anode, $v_p = \omega p / \phi$ (p = cavity pitch), and the beam drift is matched to it. That is the crossed-field amplifier of §5.6.
- Write the sketch with the cavities marked $+$ / $-$ according to ϕ and draw the spokes in the retarding cavities; that sketch carries most of the marks.
- **Uses**: radar transmitters, microwave ovens (2.45 GHz), industrial heating, linear accelerators.
- verified/added **Frequency pushing**: a change in anode voltage changes the electron drift speed and so the oscillation frequency; cure, a regulated supply.
- verified/added **Frequency pulling**: a change in load impedance (antenna VSWR) reflects into the cavities and shifts the frequency; cure, an isolator or circulator between tube and load.
- verified/added **Pros**: high peak power (MW pulsed) from a small, efficient tube. **Cons**: noisy, poor frequency stability and control, so it is an oscillator, not a precision source.

5.6 Crossed-Field Amplifier

Explain the working principle of a circular-beam type microwave cavity-controlled crossed-field amplifier 3+7 7

PIN 2/24 72 Ma, 82 Bh · the magnetron turned into an amplifier

- A **crossed-field amplifier (CFA)**, also amplatron or platinotron: magnetron geometry, $E \perp B$, but with an **RF input** and an **RF output**.
- **Circular-beam (re-entrant beam)**: the electron stream circulates round the cathode continuously, as in a magnetron.
- **Cavity-controlled (non-re-entrant RF circuit)**: the anode cavities form a slow-wave line from input to output; between output and input sits an **RF absorber / insulator**, so the wave cannot circulate and the tube does not oscillate.
- **Working**:
 - The input wave travels along the anode at $v_p = \omega p / \phi$.
 - Electrons drift at E/B ; matched to v_p they **bunch into spokes** in the retarding phase of the wave (§5.5).
 - Spokes move with the wave and give up energy cavity by cavity → the wave grows → amplified output.
 - Left-over electrons carry little RF past the absorber and re-enter the input region.
- **Performance**: high efficiency (up to about 70%), broad band (about 10%), **low gain** (10 to 15 dB), high power.
- **Use**: final power stage of radar transmitters, driven by a TWT.



Circular CFA: input, output, absorber, spiralling electron flow. Student presentation 2078.

5.7 Travelling Wave Tube

Describe the construction and operation of a travelling wave tube (TWT) 2+6

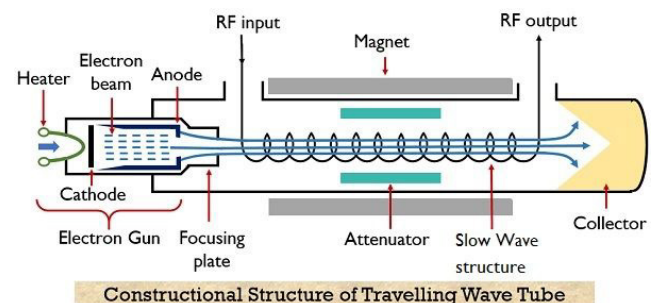
PIN 2/24 79 Bh, 80 Bh

Construction

- **Electron gun**: cathode, focusing plate, anode.
- **Helix**: a wire coil round the beam, the **slow-wave structure**; RF in at the gun end, out at the collector end.
- **Attenuator** midway along the helix: absorbs reflected waves so the tube cannot oscillate.
- **Focusing magnets** (solenoid or periodic permanent magnets) along the tube.
- **Collector**: collects the spent beam.
- **No resonant cavities** → very wide bandwidth.

Operation

1. RF on the helix travels round the turns at about c , but **along the axis** only at $v_p \approx c \frac{\text{pitch}}{2\pi r}$ (a few percent of c).
2. The beam voltage is set so the electrons move slightly **faster** than v_p .
3. The axial RF field velocity-modulates the beam → bunches form, and because beam and wave travel together the interaction lasts the **whole length**.
4. Being slightly faster, the bunches sit in the retarding field and give energy to the wave, which grows exponentially → gain 30 to 60 dB.



TWT construction. Student presentation 2078.

- **Correction to the source** (PDF p38): the beam does **not** move at the velocity of light; the whole point of the helix is to slow the wave down to the beam. Example (`tubes.py`): pitch 1 mm, radius 2 mm gives $v_p = 0.079c$, matched by a beam of about 1.6 kV; a 1 kV beam moves at $0.063c$. verified/added
- **Performance:** 0.5 to 95 GHz, octave bandwidth, 5 to 20% efficiency, mW to 10 MW pulsed. **Uses:** satellite transponders (long life), broadband receivers, radar, troposcatter links, EW jammers.

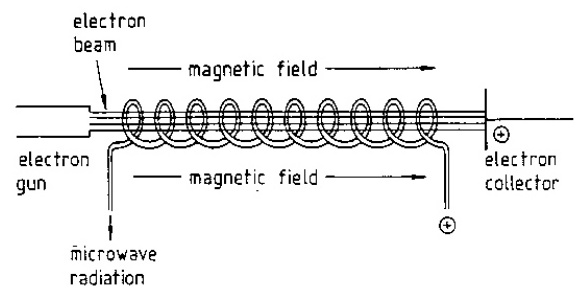
	Two-cavity klystron	TWT
Interaction	at two gaps only	continuous, whole helix
RF circuit	resonant cavities	non-resonant slow-wave helix
Bandwidth	narrow (a few %)	wide (an octave)
Wave	standing wave in cavities	travelling wave
Oscillation risk	needs feedback to oscillate	attenuator needed to prevent it

5.8 Backward Wave Oscillator and Slow-Wave Structures

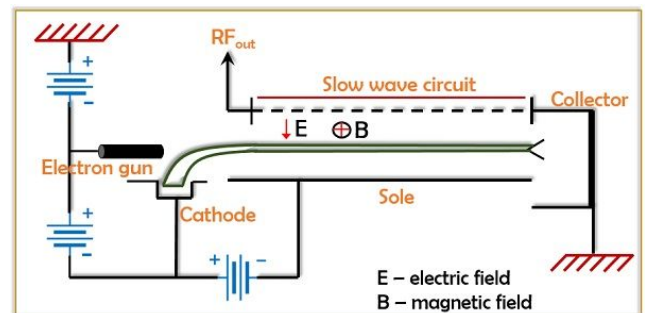
What is bunching effect? Explain the working principle of a BWO with neat diagrams 5 2+8

HOT 3/24 71 Bh, 73 Ma, 74 Bh · 71 Bh and 74 Bh as short notes

- **BWO** (carcinotron): a TWT-family tube used as an **oscillator**, microwave up to terahertz.
- Electron gun, slow-wave structure (helix or folded line), collector, magnetic focusing. **No RF input.**
- The beam interacts with a **backward space harmonic**: its phase velocity is along the beam (so they stay in step) but its **group velocity**, the direction energy flows, is **towards the gun**.
- Bunches give energy to this wave; the energy travels back to the gun end, where it velocity-modulates newly arriving electrons: **built-in feedback** → self-sustained oscillation.
- **Output is taken at the gun end**; the collector end is matched.
- **Frequency is tuned electronically** by the beam voltage (it sets which backward harmonic is in step): very wide tuning range.
- M-type BWO: crossed fields with a negative **sole** electrode (lower figure).
- Uses: swept signal generators, THz spectroscopy and imaging, jammers.
- **Not** a klystron: some student notes describe buncher and catcher cavities and a feedback conductor; a BWO has none.



O-type BWO: output near the gun. Deck slide 40.



Schematic representation of Backward Wave Oscillator

Electronics Desk

M-type BWO with sole. Student presentation 2078.

What do you mean by a slow (backward) wave structure? 2

73 Bh

- A **slow-wave structure** is a periodic RF circuit that makes the wave's axial phase velocity much smaller than c , so an electron beam of practical voltage can keep in step with it: helix, folded-back line, ring-bar, interdigital line, coupled cavities.
- It is **backward** when the space harmonic used has phase and group velocity in **opposite** directions: energy flows against the beam, as the BWO needs.

5.9 Low-Noise Amplifier

Explain the construction and working principle of an LNA 2+6

PIN 1/24 73 Bh · verified/added no source note covers it; amplifier design in Ch 6

- **Purpose:** the first amplifier after the antenna. Its noise figure sets the whole receiver's, by Friis:

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} \quad (\text{one more term per stage})$$

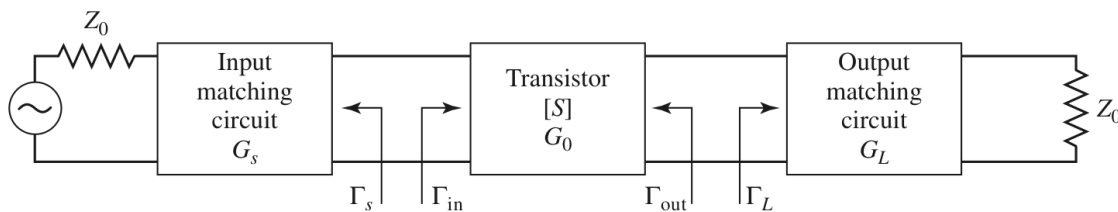
e.g. a 1 dB, 20 dB-gain LNA before an 8 dB mixer gives 1.18 dB overall; the mixer first would give 8.02 dB (`tubes.py`).

- **Construction:**

- **Low-noise device:** GaAs FET / HEMT (historically a parametric amplifier, maser or low-noise TWT).
- **Input matching network:** presents the source reflection Γ_{opt} for **minimum noise figure**, not conjugate match.
- **Output matching network:** conjugate match for gain.
- **Bias network** with RF chokes and bypass capacitors; stabilising resistors so $K > 1$.

- **Working:** the signal is amplified with gain 10 to 20 dB while adding as little noise as possible (NF 0.5 to 2 dB); trade-off between minimum NF and maximum gain drawn as noise circles and gain circles on the Smith chart (Ch 6).

The construction, drawn



Source, **input matching network** (Γ_s , set to Γ_{opt} for an LNA), transistor $[S]$, **output matching network** (Γ_L , conjugate for gain), load. Every amplifier in Ch 6 is this block diagram. Pozar Fig 12.2.

5.10 PYQ Mapping

Theory questions, covered in this document

- **Limitations of conventional tubes; transit-time effect** → §5.1. 70 Ma 2, 77 Ch 2, 79 Bh 2
- **Bunching effect / density modulation** → §5.2. **Most repeated ask of the chapter.** 70 Bh 2, 71 Bh 2, 72 Ash 2, 73 Ma 2, 80 Ba 2, 80 Bh 2, 81 Bh 2
- **Velocity modulation and bunching in a (multicavity) reflex klystron** → §5.2 and §5.4. 72 Ma 5, 75 Bh 10
- **Two-cavity klystron amplifier** → §5.3. 69 Bh 5, 70 Ma 8
- **Multicavity klystron amplifier / oscillator; klystron oscillator** → §5.3. 71 Bh 8, 72 Ash 7, 78 Ch 8, 80 Ch 5, 82 Ba 4
- **Cross-field effect** → §5.5. 72 Ma 3, 78 Ch 2, 81 Bh 2
- **Cavity magnetron; oscillator using bunching and field effects** → §5.5. 70 Bh 8, 71 Ma 10, 79 Ch 5, 81 Bh 4
- **Magnetron with 45 / 60 / 90 degree phase shift** → §5.5. 74 Ma 8, 77 Ch 6, 80 Ba 6, 81 Ba 8, 72 Ma 7
- **Circular-beam cavity-controlled crossed-field amplifier** → §5.6. 82 Bh 7
- **TWT construction and operation** → §5.7. 79 Bh 6, 80 Bh 6
- **Backward wave oscillator** → §5.8. 71 Bh 5, 73 Ma 8, 74 Bh 5
- **Slow backward wave structure; LNA** → §5.8 and §5.9. 73 Bh 2+6

Numerical content

- **This chapter has no numerical component:** no paper in 24 sets a tube calculation. The formulas above (bunching parameter, Hull cut-off, reflex transit condition, helix velocity) are there to strengthen descriptive answers, and each is checked in `src/tubes.py`.

Corrections, and fixes made to the PYQ documents while building this chapter

- Source PDF p23: a klystron catcher converts only part of the beam energy (58% theoretical maximum), not all of it.
- Source PDF p38: a TWT beam moves far slower than light.
- Deck slide 27: reflex efficiency 22.78% is X' rounded; exact 22.7%.

- Student BWO slides describe klystron cavities; ignored.
- Sorted PYQ documents: 71 Bh's BWO short note was missing; 73 Ma is "bunching + BWO" and 73 Bh "slow backward wave structure + LNA"; 70 Ma asks no bunching question.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.